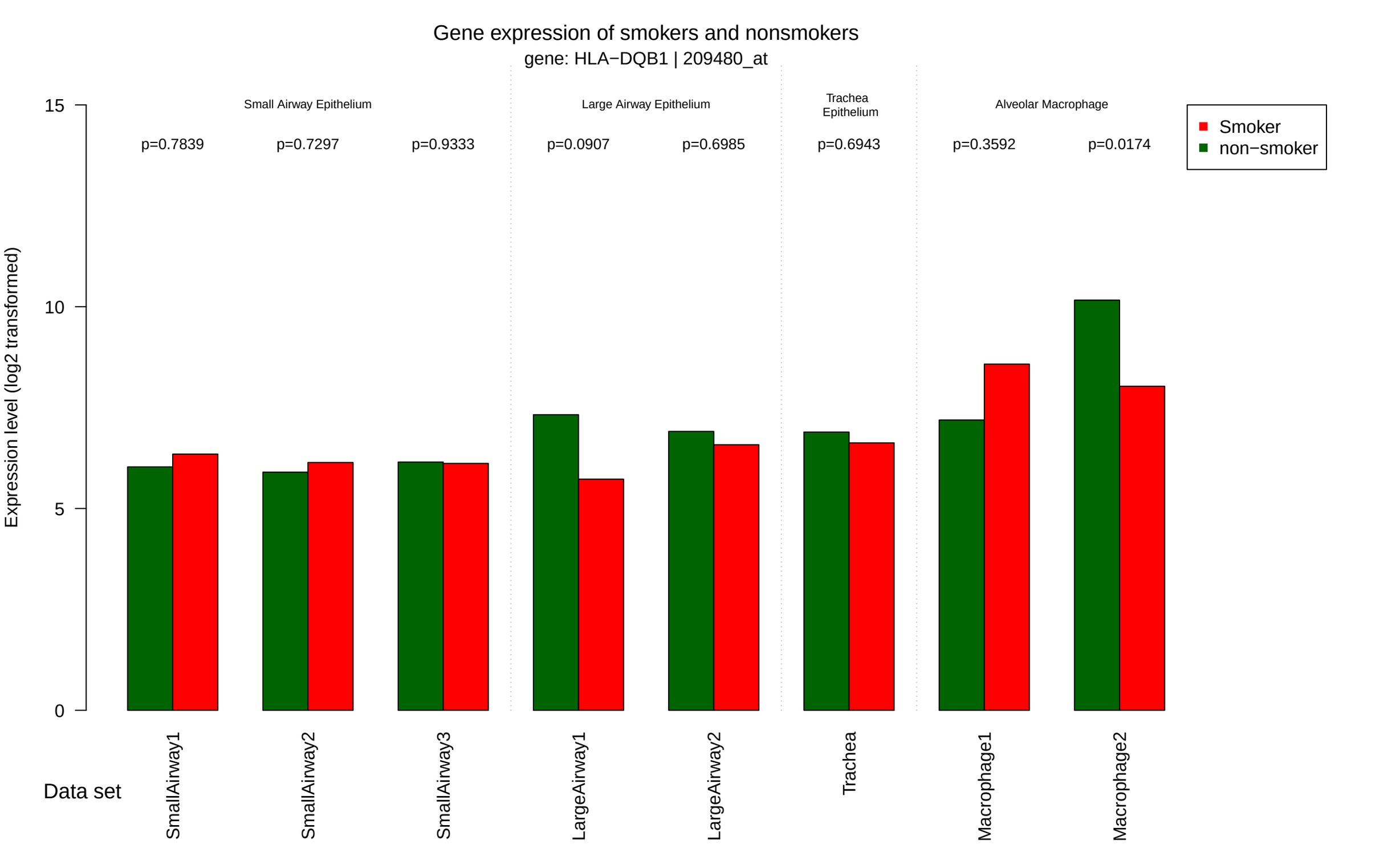
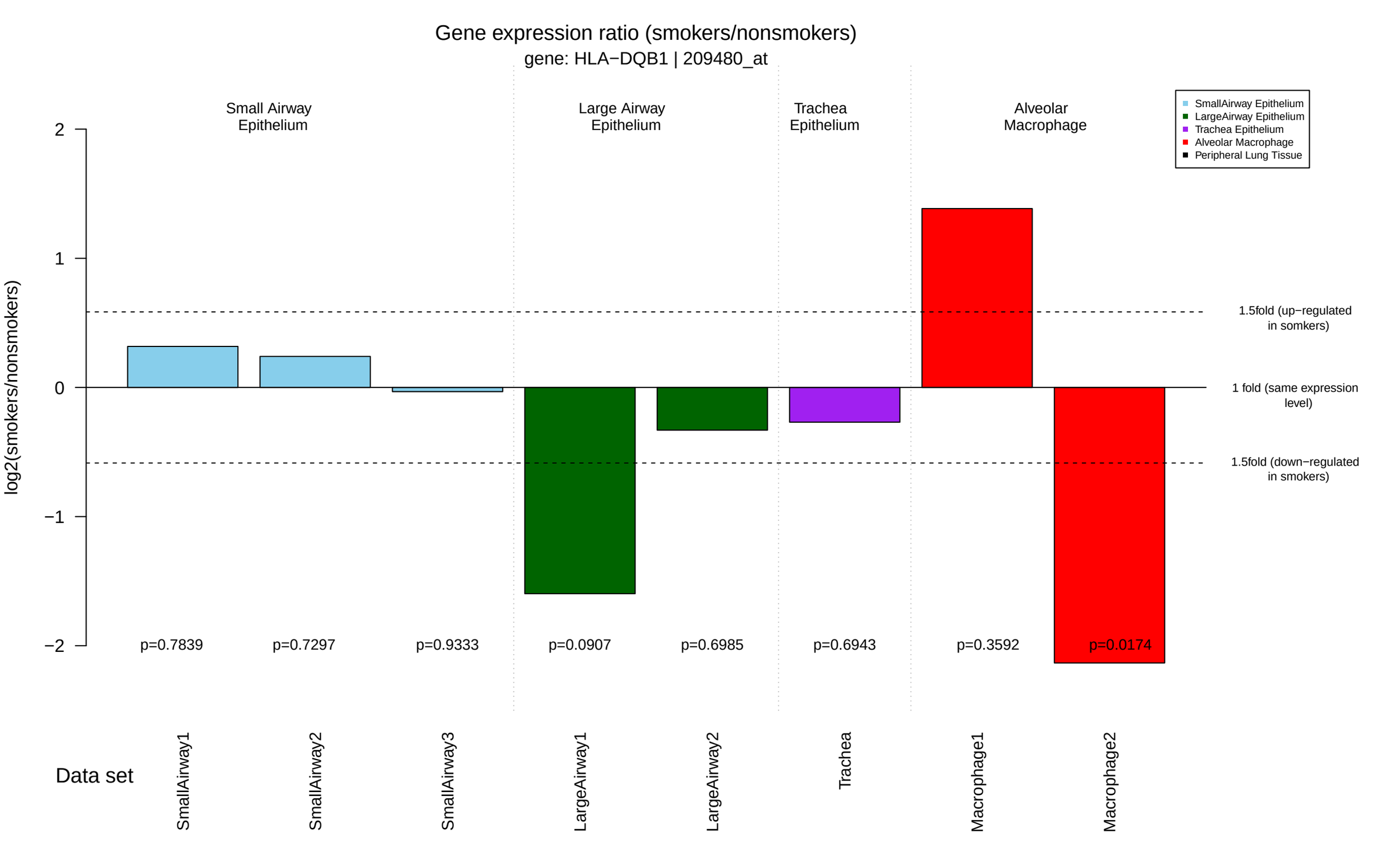
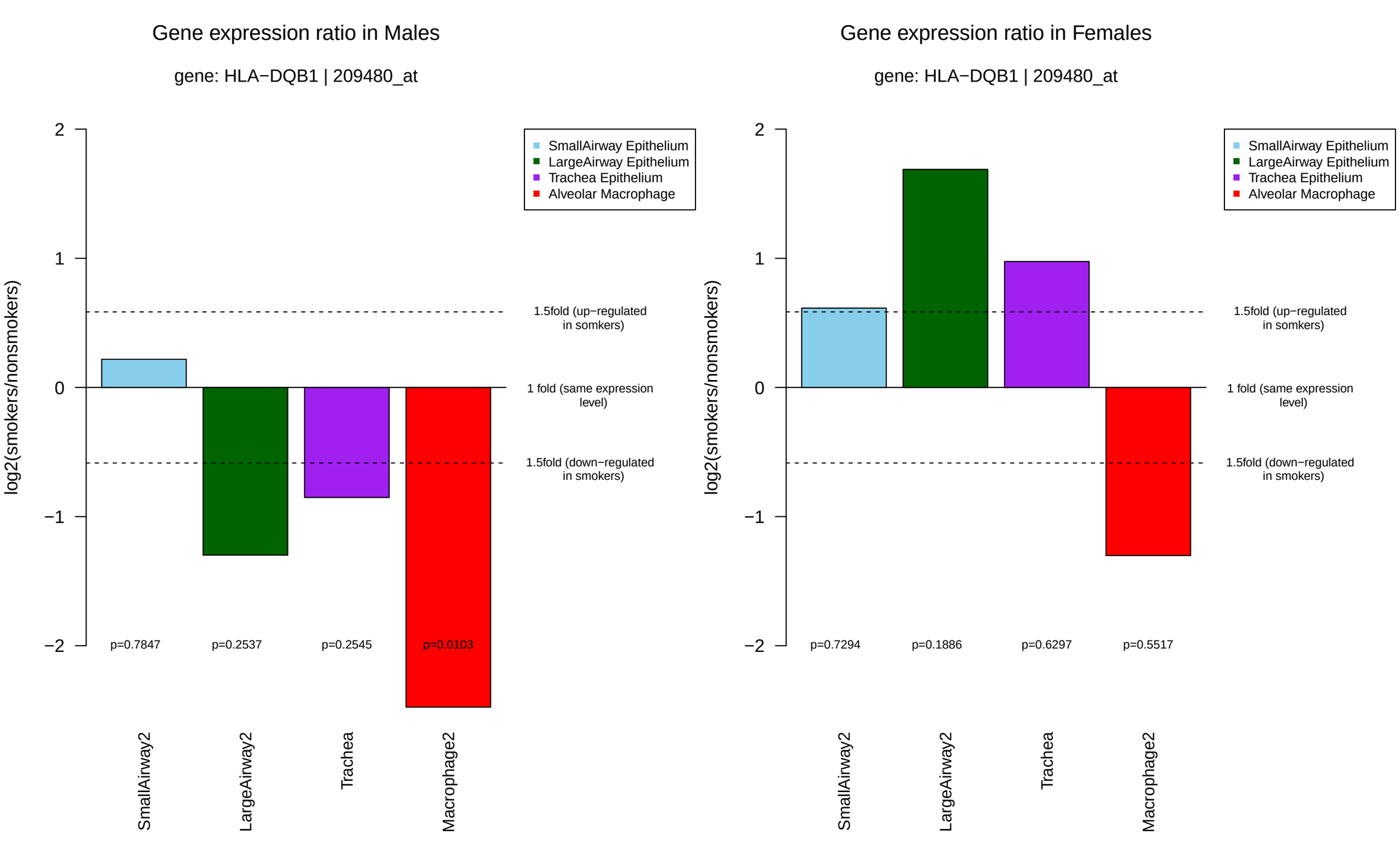
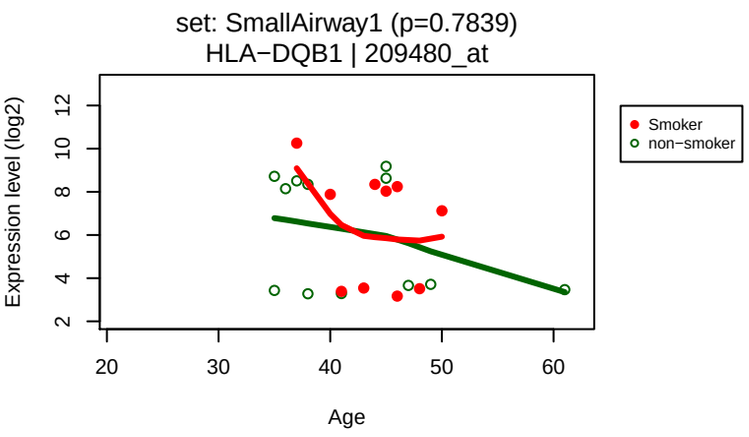
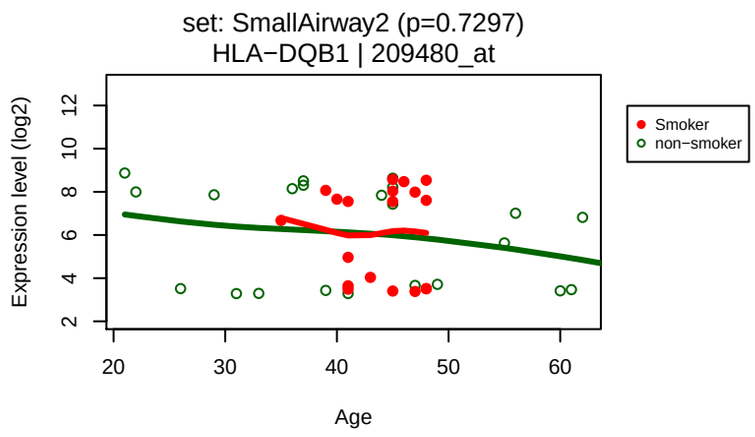
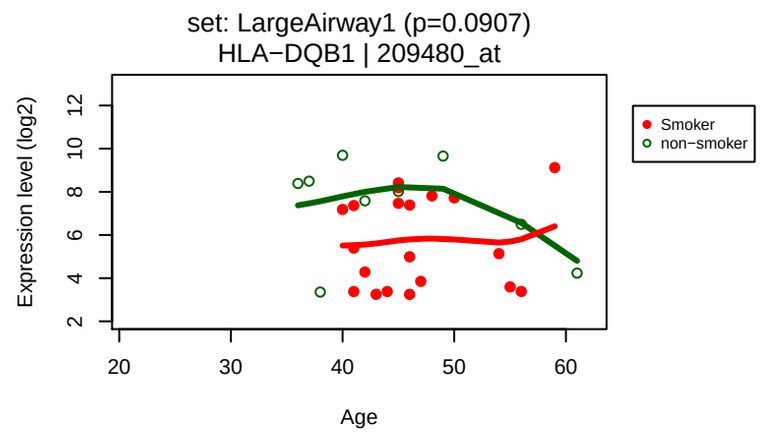

Gene expression of smokers and nonsmokers
gene: HLA−DQB1 | 209480_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7839
p=0.7297
p=0.9333
p=0.0907
p=0.6985
p=0.6943
p=0.3592
p=0.0174
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

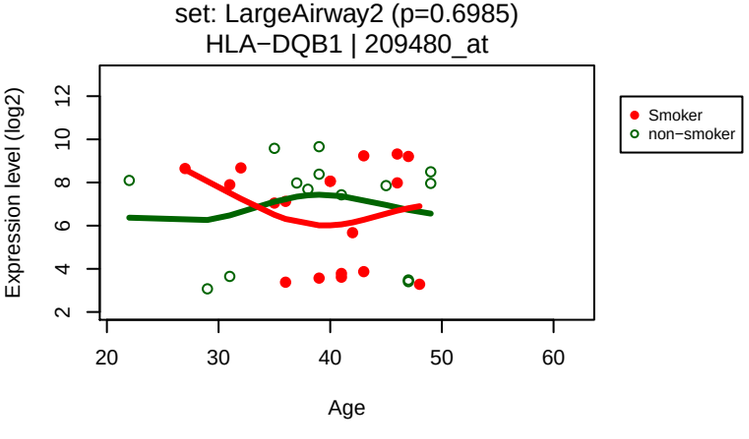

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DQB1 | 209480_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7839
p=0.7297
p=0.9333
p=0.0907
p=0.6985
p=0.6943
p=0.3592
p=0.0174
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

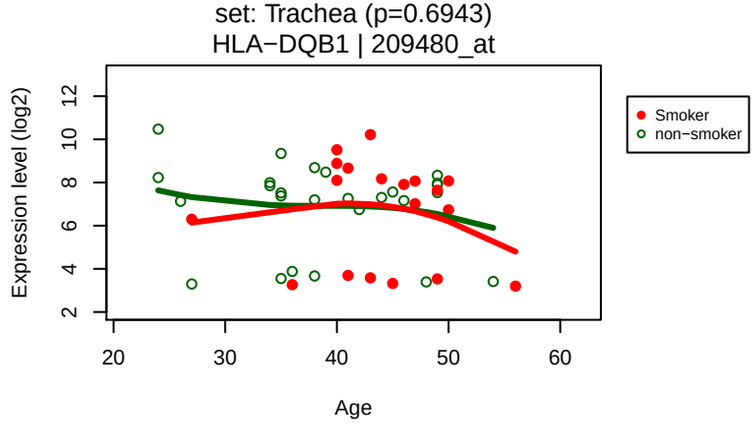

Gene expression ratio in Males
gene: HLA−DQB1 | 209480_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7847
p=0.2537
p=0.2545
p=0.0103
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DQB1 | 209480_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7294
p=0.1886
p=0.6297
p=0.5517
SmallAirway2
LargeAirway2
Trachea
Macrophage2

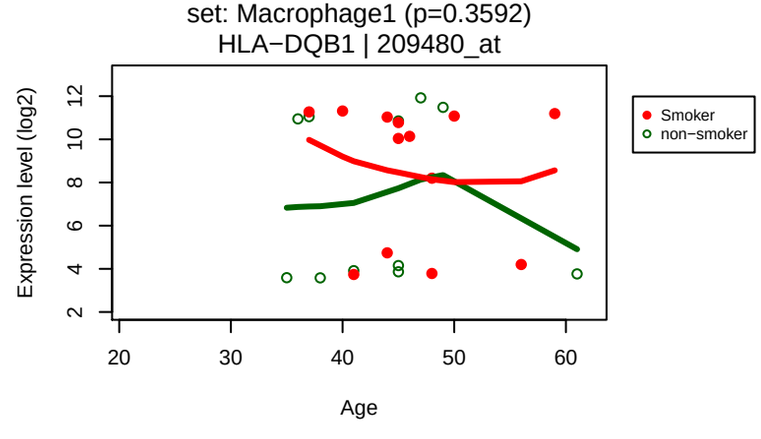

set: SmallAirway1 (p=0.7839)
HLA−DQB1 | 209480_at
Expression level (log2)
Age
Smoker
non−smoker

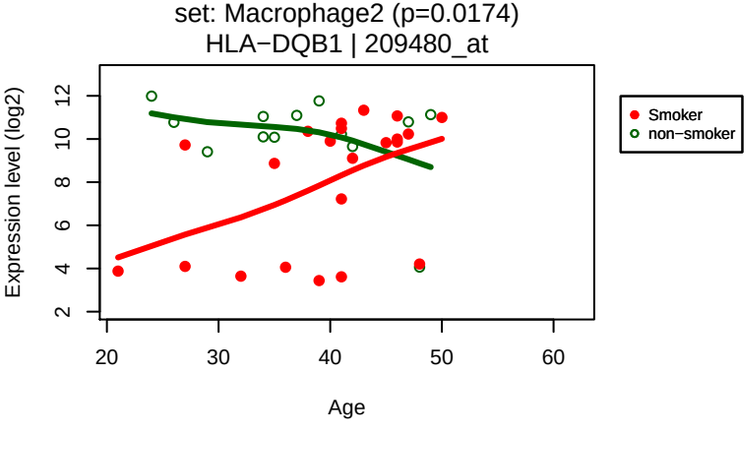

set: SmallAirway2 (p=0.7297)
HLA−DQB1 | 209480_at
Expression level (log2)
Age
Smoker
non−smoker

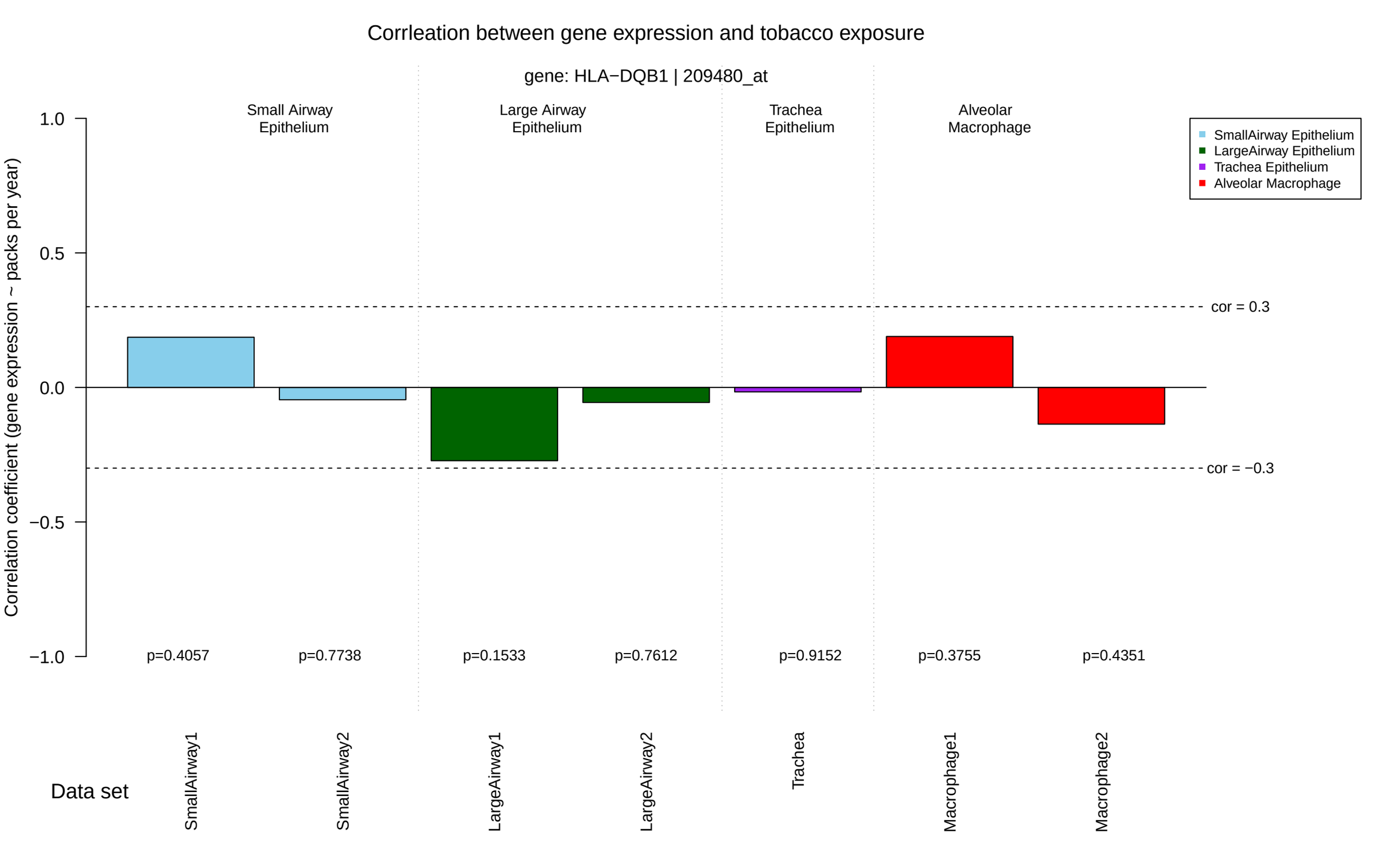

set: LargeAirway1 (p=0.0907)
HLA−DQB1 | 209480_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6985)
HLA−DQB1 | 209480_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6943)
HLA−DQB1 | 209480_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3592)
HLA−DQB1 | 209480_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0174)
HLA−DQB1 | 209480_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DQB1 | 209480_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4057
p=0.7738
p=0.1533
p=0.7612
p=0.9152
p=0.3755
p=0.4351
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2